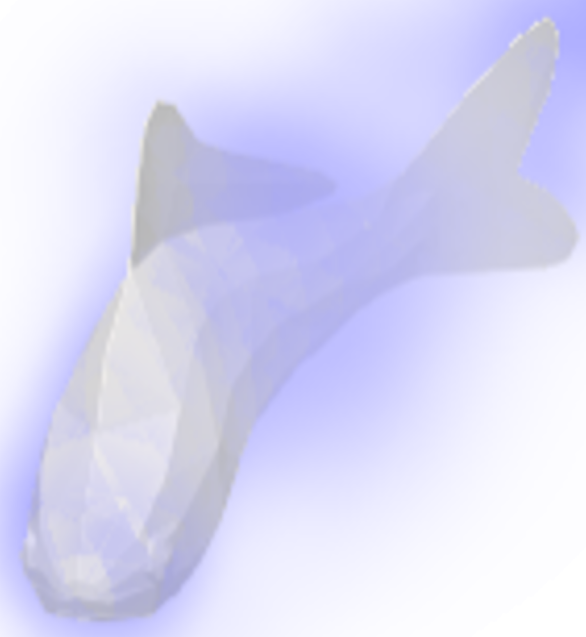




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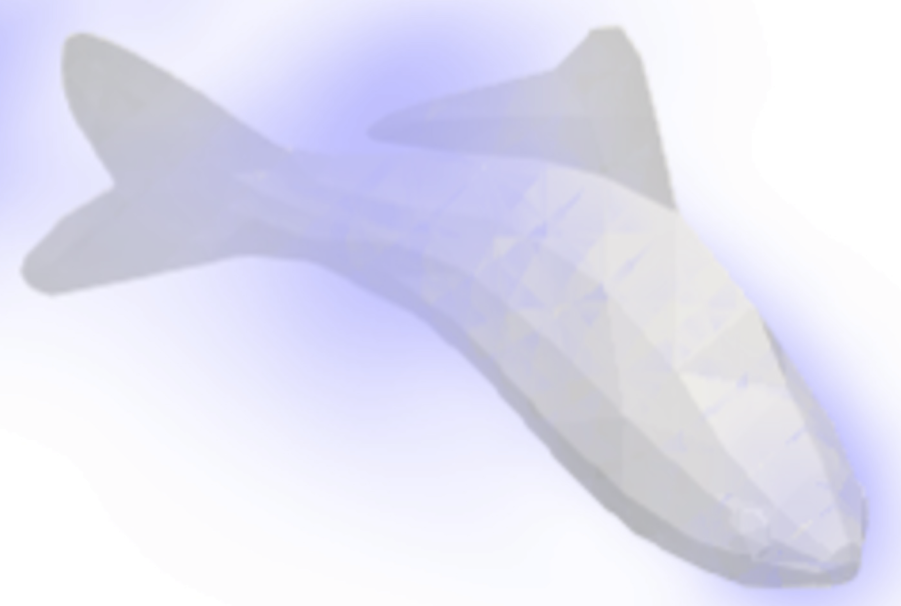


Agile and Controllable Omnidirectional Fast-Start Maneuvers of Robotic Fish Via Bio-inspired Reinforcement Learning

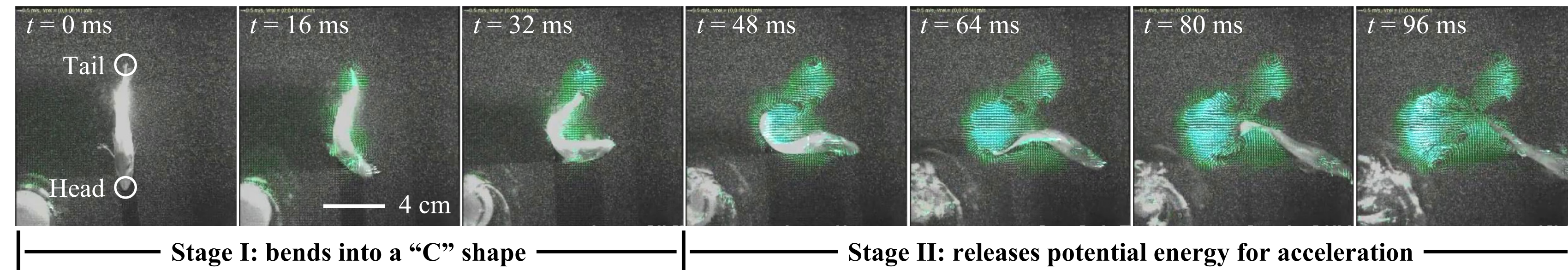
Xu Huang*, Xiaozhu Lin*, Xiaopei Liu, and Yang Wang
ShanghaiTech University



Video



Motivation



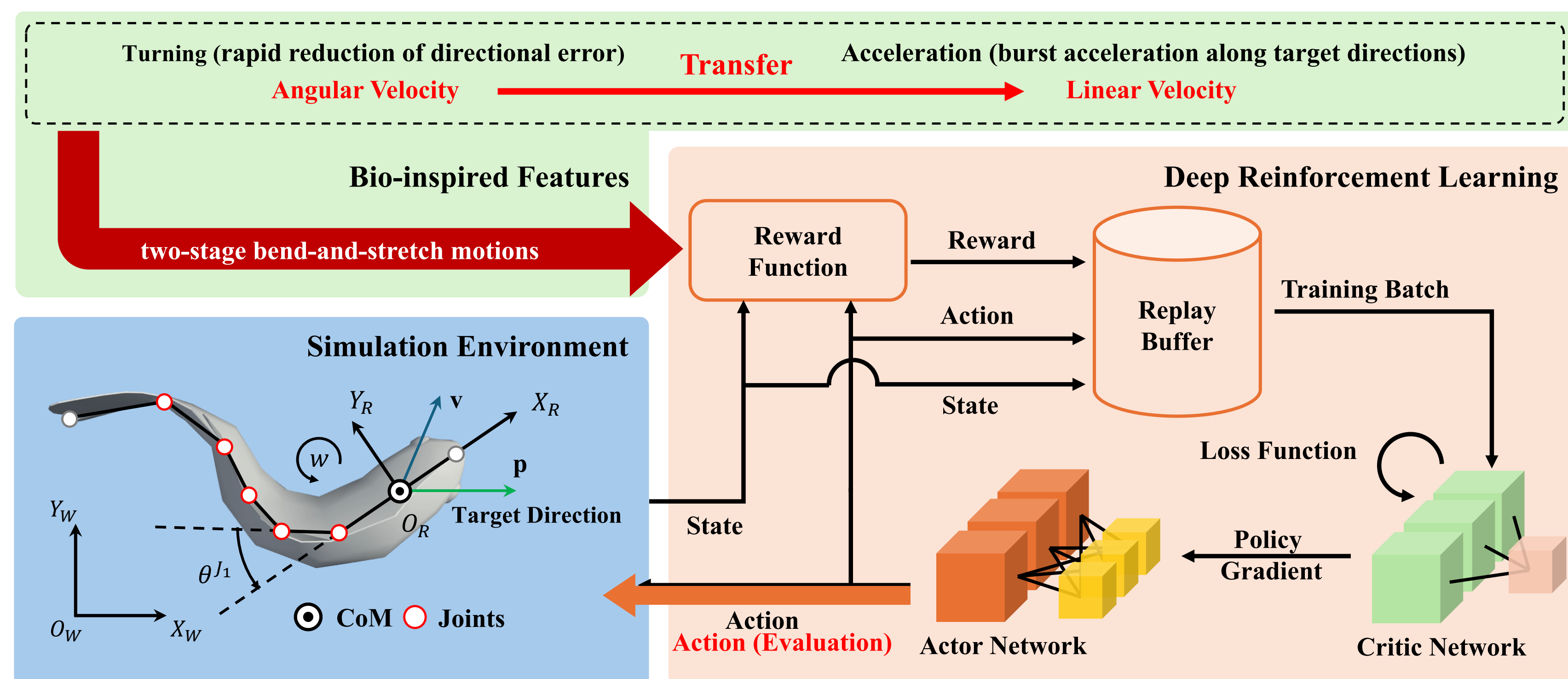
Background

Prey fish often employ high-acceleration, explosive maneuvers, *i.e.*, fast-starts, when escaping from predators or attacking prey. Reproducing fast-start maneuvers using robotic systems is essential for advancing understanding of fish locomotion, yet it poses significant challenges in terms of materials, actuation, perception, and **control**.

Shortcomings of Existing Works

- ✗ **Ideal Simulated Model:** Ideal simulated model (e.g., continuum) fail to generalize to real-world robotic fish (usually articulated rigid body).
- ✗ **Only Turning Stage:** Focus on rapid turning, but insufficient forward acceleration.
- ✗ **Rely on Passive Materials:** Relying on specialized materials for passive ejection acceleration instead of controllable policy.

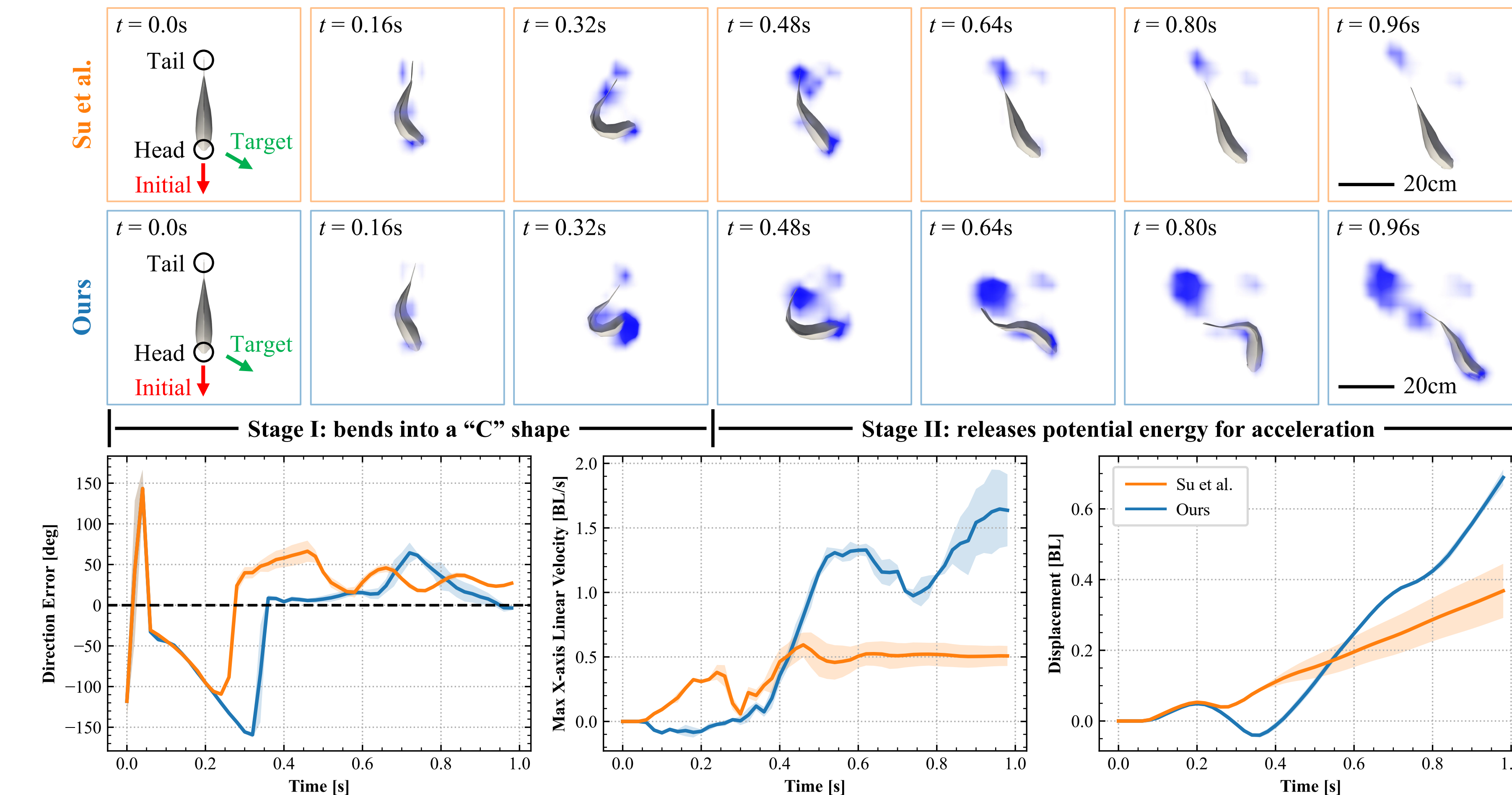
Bio-inspired DRL Framework



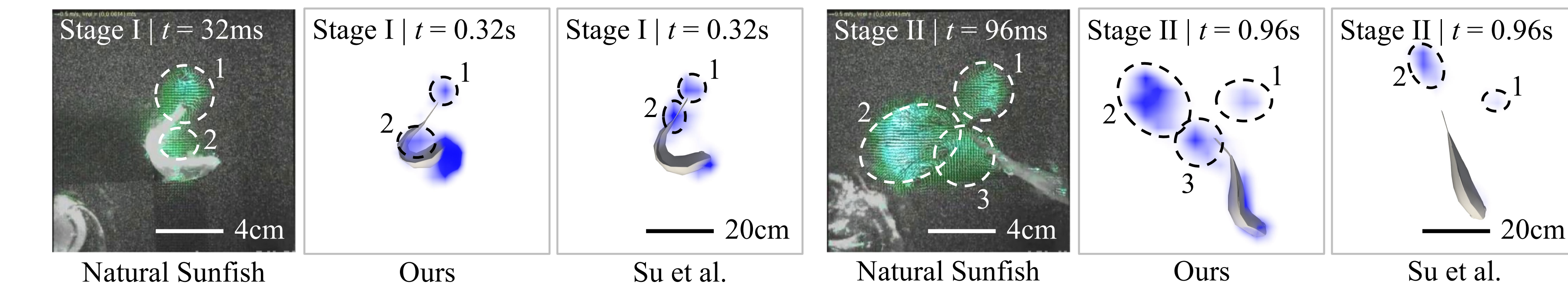
- ✓ **Observation Space:** 17-dimensional observation vector capturing robot kinematics, joint states, and stage phase flag, $\mathcal{S}_t = [\mathbf{p}, \boldsymbol{\omega}, \mathbf{v}, \mathbf{J}, \mathbf{j}, \tau_t, \sigma_t]$.
- ✓ **Action Space:** Direct set as desired joint positions without CPG or pre-design pattern, which enables to explore non-periodic and agile policy, $\mathcal{A}_t = [\mathbf{J}_{des}] \in \mathbb{R}^5$.
- ✓ **Bio-inspired Reward Design:** A coupled, stage-specific objective function by biological fast-start phase, $R = \sum_{t=0}^T (r_t^{\text{flag}} + r_t^{\text{penalty}}) + \omega_{\text{prop}}(X_T - X_0) * \mathbf{p}$.
- ✓ **Schedule Detail:** Control timestep is 0.02 s, robot position and orientation randomization, and target direction randomization.

Results

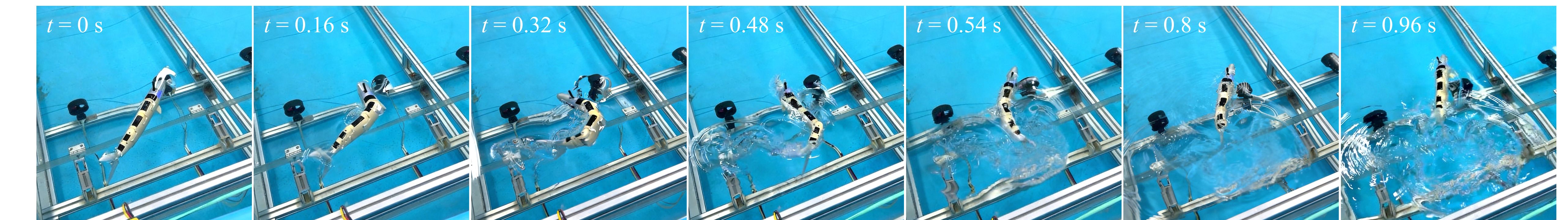
Exp.1 / 60° Directional Maneuver



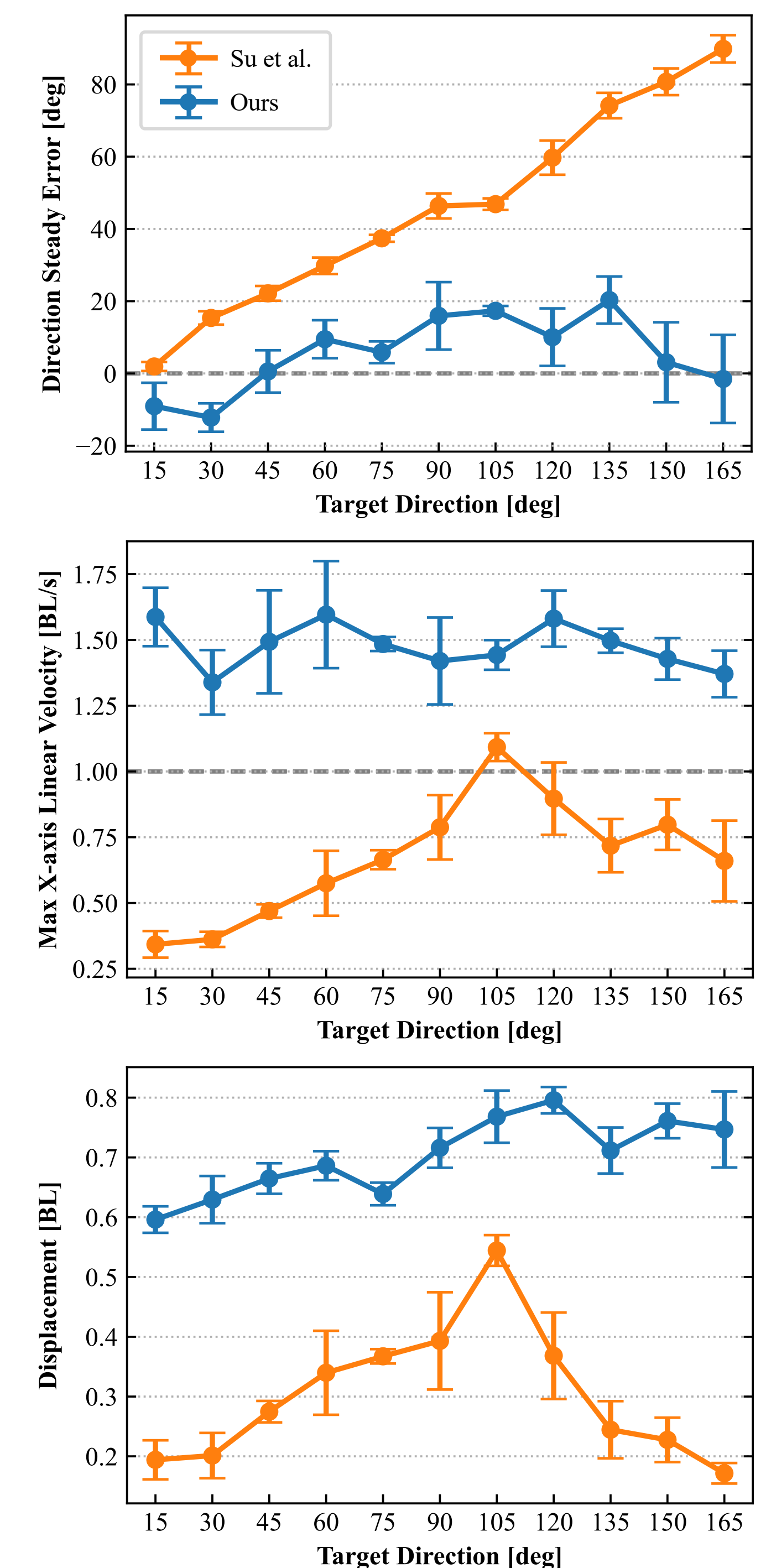
Exp.3 / Vortex Structures Comparison



Exp.4 / Sim-to-Real Transfer

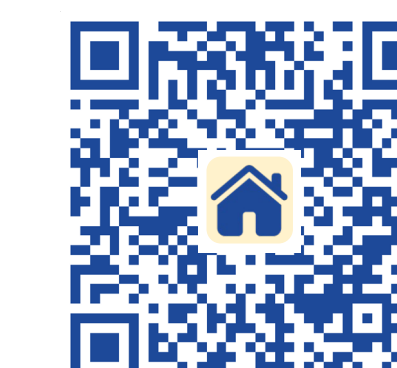


Exp.2 / Omnidirectional Analysis



Conclusion & Future Work

- **RL for Non-periodic Agile Maneuvers:** This work demonstrated that the DRL-based control enable to effectively synthesize transient, non-periodic maneuvers for multi-joint robotic fish, such as adaptive *fast-start* escape behaviors in this work.
- **From Bio-inspired Observation to Reward Design:** This work also proved that integrating a biological two-stage reward with a high-performance CFD simulator yields superior directional accuracy and efficient forward propulsion over baselines.
- **Complex Fluid Scenario:** Generalizing the policy to more complex fluid scenarios, such as flow disturbances, even wild field.
- **Hardware Optimization:** Explore the co-design and benefits of advancing smart materials and flexible structural mechanics.



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